ISLAMIC UNIVERSITY OF TECHNOLOGY (IUT)

ORGANISATION OF ISLAMIC COOPERATION (OIC)

Department of Computer Science and Engineering (CSE)

QUIZ-1 SET-A **SUMMER SEMESTER, 2023-2024**

DURATION: 30 MINUTES **FULL MARKS: 15**

Math 4341: Linear Alegbra

Programmable calculators are not allowed. Write your Student ID at the top of your extra pages (if any).

Answer **all 4 (four)** questions. Figures in the right margin indicate full marks of questions whereas corresponding CO and PO are written within parentheses.

Student ID:

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1. You have been provided with three vectors: 2 (CO1) (PO1)

$$\mathbf{a} = \begin{bmatrix} 1 \\ 1 \\ -1 \end{bmatrix}, \quad \mathbf{b} = \begin{bmatrix} 2 \\ 3 \\ 4 \end{bmatrix}, \quad \mathbf{c} = \begin{bmatrix} 3 \\ 4 \\ 3 \end{bmatrix}.$$

And vector

$$\mathbf{d} = \mathbf{a} + \mathbf{b} + \mathbf{kc}$$

where k is a scalar. For what value of k will the span of $\mathbf{d}$ cover the entire 3D space?

Solution:

$$a + b = c \tag{1}$$

or,

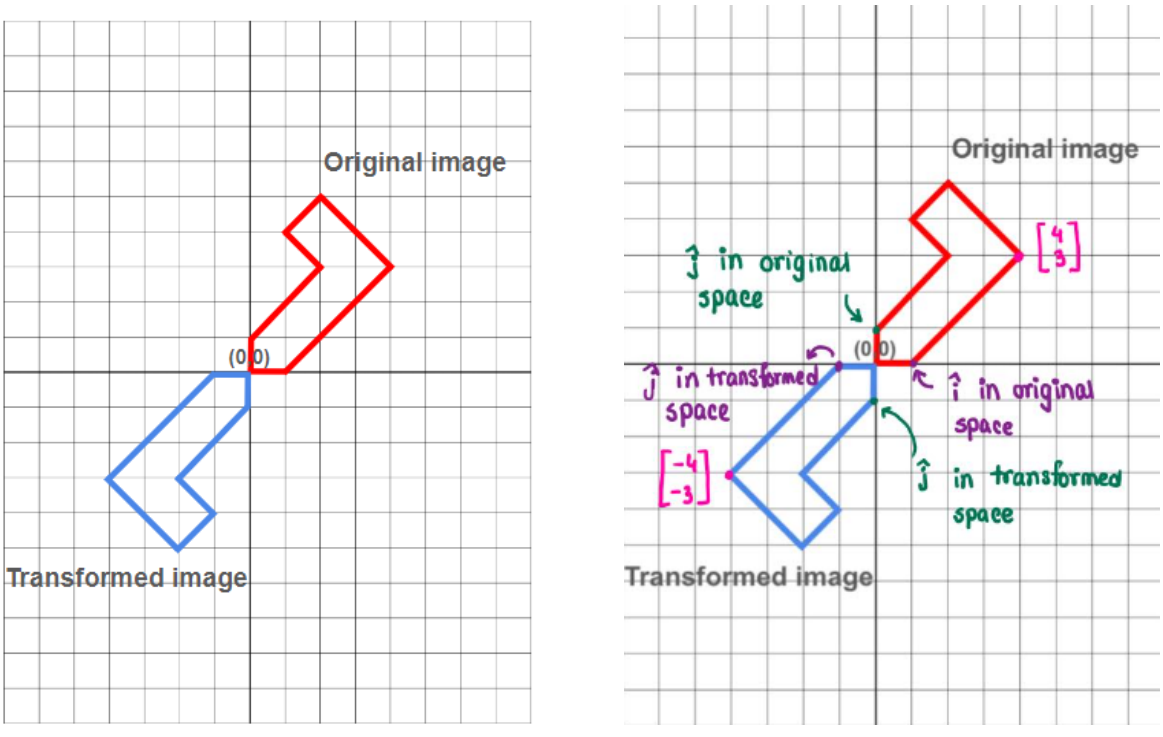
$$\begin{bmatrix} 1 \\ 1 \\ -1 \end{bmatrix} + \begin{bmatrix} 2 \\ 3 \\ 4 \end{bmatrix} = \begin{bmatrix} 3 \\ 4 \\ 3 \end{bmatrix} \tag{2}$$

So, c is the linear combination of a and b (vector addition to be precise). k is just the scaling factor. So, no matter what the value of k is, the span is just a straight line (scaled vector c or (a+b)).

Rubric:

- 1 point for pointing out that c is the linear combination of a and b.
- 1 point for any comment on the span (straight line or scaled c or (a+b) vector).

2. The basis vectors for the original image in Figure 1 are $\hat{\mathbf{i}} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$ and $\hat{\mathbf{j}} = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$. Determine the basis vector for the transformed image and **verify** that you have correctly determined the basis vector of the transformed space with another vector other than the basis vectors. (**Note:** Each square on the graph paper represents one unit in length.) 3 (CO1) (PO1)



(a) Figure from the question

(b) Solution

Figure 1

Solution:

If we map the original image with the transformed image, you'll be able to see that $\hat{\mathbf{i}}$ falls on $\begin{bmatrix}-1\\0\end{bmatrix}$ and $\hat{\mathbf{j}}$ falls on $\begin{bmatrix}0\\-1\end{bmatrix}$.

So, basis vector of original space, $\hat{\mathbf{i}} = \begin{bmatrix}1\\0\end{bmatrix}$ and $\hat{\mathbf{j}} = \begin{bmatrix}0\\1\end{bmatrix}$.

Basis vector of the transformed space , $\hat{\mathbf{i}} = \begin{bmatrix}-1\\0\end{bmatrix}$ and $\hat{\mathbf{j}} = \begin{bmatrix}0\\-1\end{bmatrix}$.

Lets take one vector as example, the outward corner can be represented as $\begin{bmatrix}4\\3\end{bmatrix}$.

Its supposed to land on $\begin{bmatrix}-4\\-3\end{bmatrix}$ after transformation.

Now, in original space,

$$\begin{bmatrix}1 & 0\\0 & 1\end{bmatrix}\begin{bmatrix}4\\3\end{bmatrix} = 4\begin{bmatrix}1\\0\end{bmatrix} + 3\begin{bmatrix}0\\1\end{bmatrix} = \begin{bmatrix}4\\3\end{bmatrix}$$

Now, in transformed space,

$$\begin{bmatrix}-1 & 0\\0 & -1\end{bmatrix}\begin{bmatrix}4\\3\end{bmatrix} = 4\begin{bmatrix}-1\\0\end{bmatrix} + 3\begin{bmatrix}0\\-1\end{bmatrix} = \begin{bmatrix}-4\\-3\end{bmatrix}$$

Rubric:

- 1 point for successfully determining each basis vector in transformed space.
- 1 for verifying a point in transformed space.

3. Given, 3 (CO1) (PO1)

$$A = \begin{bmatrix} 2 & -3 & 8 \\ -9 & 2 & -7 \\ 5 & 7 & 1 \end{bmatrix}$$

a) What is the permutation matrix that exchanges row 1 with row 2?

b) What is the permutation matrix that exchanges column 1 with column 2?

c) Are the two permutation matrices from 3(a) and 3(b) the same? If so, how is the same matrix responsible for both row and column exchange?

Solution:

a) $P_{12} = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}$

b) $P_{12} = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}$

c) Yes. They are the same. But they need to be multiplied with desired matrix in different order.

For row swap, $P_{12}A = RowExchangedMatrix$

$$\begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 2 & -3 & 8 \\ -9 & 2 & -7 \\ 5 & 7 & 1 \end{bmatrix} = \begin{bmatrix} -9 & 2 & -7 \\ 2 & -3 & 8 \\ 5 & 7 & 1 \end{bmatrix}$$

For column swap, $AP_{12} = ColumnExchangedMatrix$

$$\begin{bmatrix} 2 & -3 & 8 \\ -9 & 2 & -7 \\ 5 & 7 & 1 \end{bmatrix} \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} -3 & 2 & 8 \\ 2 & -9 & -7 \\ 7 & 5 & 1 \end{bmatrix}$$

Rubric:

- 1 points for getting each permutation matrix right.
- 1 points for getting the order of multiplication right.

4. You are provided with the following equations - $1 + 0.5 + 4.5 + 1$ (CO2) (PO1)

$$2x + y + 5z = -1 \quad (3)$$

$$6x + 3y + 7z = 5 \quad (4)$$

$$4x + 5y + 3z = 11 \quad (5)$$

a) Write down the system of linear equations in $Ax = b$ or Matrix form.

b) Write down the augmented matrix.

c) Perform necessary operations on the coefficient matrix A to obtain the upper triangular matrix U using elimination matrices E_{ij}.
Convert $\mathbf{Ax} = \mathbf{b}$ to $\mathbf{Ux} = \mathbf{c}$ form. (Where U is an upper triangular matrix.)

d) Find out the values of x,y and z using back substitution.

Solution:

It's ideal to swap row 2 and 3 before starting elimination to prevent temporary failure. If you don't swap them, that's fine too, just keep in mind you'll need to use a permutation matrix later.

$$2x + y + 5z = -1 \tag{6}$$
$$4x + 5y + 3z = 11 \tag{7}$$
$$6x + 3y + 7z = 5 \tag{8}$$

a)

$$\underset{\text{A}}{\begin{bmatrix} 2 & 1 & 5 \\ 4 & 5 & 3 \\ 6 & 3 & 7 \end{bmatrix}} \underset{\text{x}}{\begin{bmatrix} x \\ y \\ z \end{bmatrix}} = \underset{\text{b}}{\begin{bmatrix} -1 \\ 11 \\ 5 \end{bmatrix}}$$

b)

$$\underset{\text{A} \quad | \quad \text{b}}{\left[\begin{array}{ccc|c} 2 & 1 & 5 & -1 \\ 4 & 5 & 3 & 11 \\ 6 & 3 & 7 & 5 \end{array}\right]}$$

c)

$$\underset{E_{21}}{\begin{bmatrix} 1 & 0 & 0 \\ -2 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}} \left[\begin{array}{ccc|c} 2 & 1 & 5 & -1 \\ 4 & 5 & 3 & 11 \\ 6 & 3 & 7 & 5 \end{array}\right] = \left[\begin{array}{ccc|c} 2 & 1 & 5 & -1 \\ 0 & 3 & -7 & 13 \\ 6 & 3 & 7 & 5 \end{array}\right]$$

$$\underset{E_{31}}{\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ -3 & 0 & 1 \end{bmatrix}} \left[\begin{array}{ccc|c} 2 & 1 & 5 & -1 \\ 0 & 3 & -7 & 13 \\ 6 & 3 & 7 & 5 \end{array}\right] = \underset{U|c}{\left[\begin{array}{ccc|c} 2 & 1 & 5 & -1 \\ 0 & 3 & -7 & 13 \\ 0 & 0 & -8 & 8 \end{array}\right]}$$

so,

$$\underset{\text{U}}{\begin{bmatrix} 2 & 1 & 5 \\ 0 & 3 & 7 \\ 0 & 0 & -8 \end{bmatrix}} \underset{\text{x}}{\begin{bmatrix} x \\ y \\ z \end{bmatrix}} = \underset{\text{c}}{\begin{bmatrix} -1 \\ 13 \\ 8 \end{bmatrix}}$$

d) which implies,

$$2x + y + 5z = -1 \tag{9}$$
$$3y + 7z = 13 \tag{10}$$

$$-8z = 8 \tag{11}$$

From, Eqn (11) we get, $z = -1$
Substituting $z = -1$ in eqn (10), we get, $3y + 7(-1) = 13$ or $y = 2$
Substituting $z = -1$ and $y = 2$ in eqn (10), we get, $2x + (2) + 5(-1) = -1$ or $x = 1$.

Rubric:

- 1 point for writing down the system of linear equations in $Ax = b$ or Matrix form.
- 0.5 point for writing down the augmented matrix.
- 1.5 each for one permutation matrix and two elimination matrix operation.
- 1 for correctly determining the values of x,y and z.

5. **BONUS** : Jawad and Utsho play the following game with a 2024×2024 matrix. Jawad fills in one of the entries of the matrix with a real number, then Utsho, then Jawad and so forth until the entire matrix is filled. At the end, if the matrix is **singular**, **Utsho wins**; otherwise, **Jawad wins**. (A player cant override a position that has already been filled.) **5** (CO204) (PO204)

   a) Who wins if both players play strategically? Provide a concise justification. [Jawad / Utsho / Depends]

   b) Does the outcome of the game depend on which player makes the last move? [Yes/No]

   c) Who makes the last move? [Jawad / Utsho]

   d) Who wins in a 2025×2025 matrix game if both players play strategically? Please provide a brief justification. [Jawad / Utsho / Depends]

Solution:

Play this game with your friend a few times and let me know who wins. Don't forget to make your move **wisely** and **strategically**.